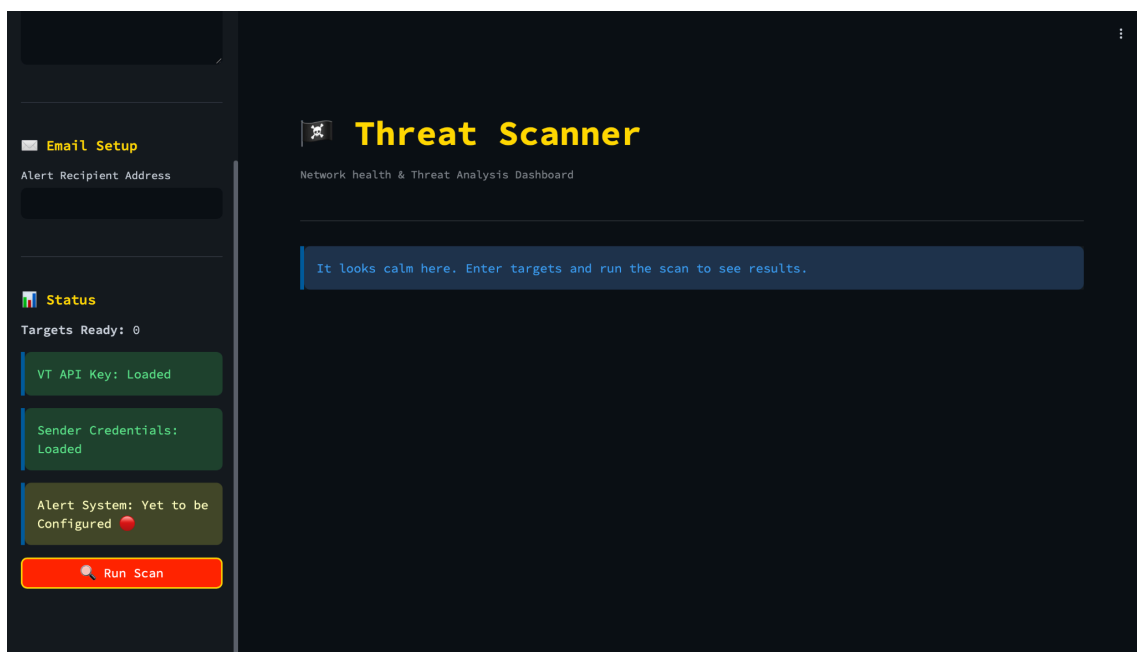


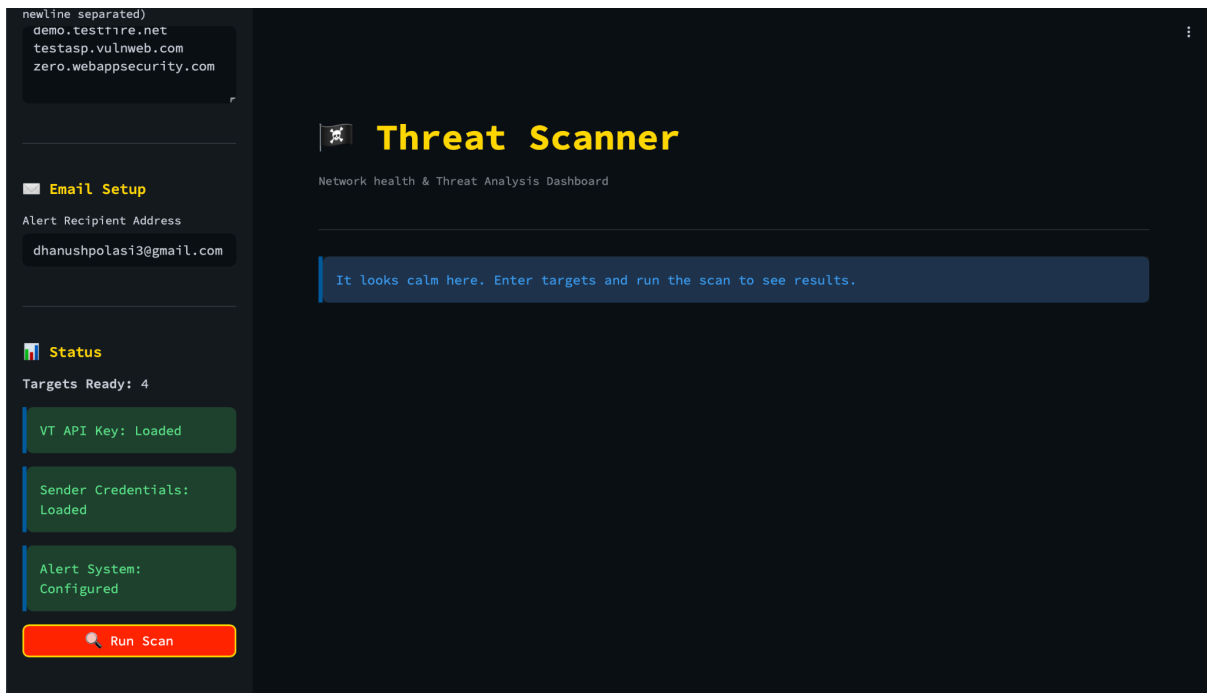
A3: Web Application Vulnerability Scanning, Risk Evaluation & Alert System

Source code : [github-link](#)

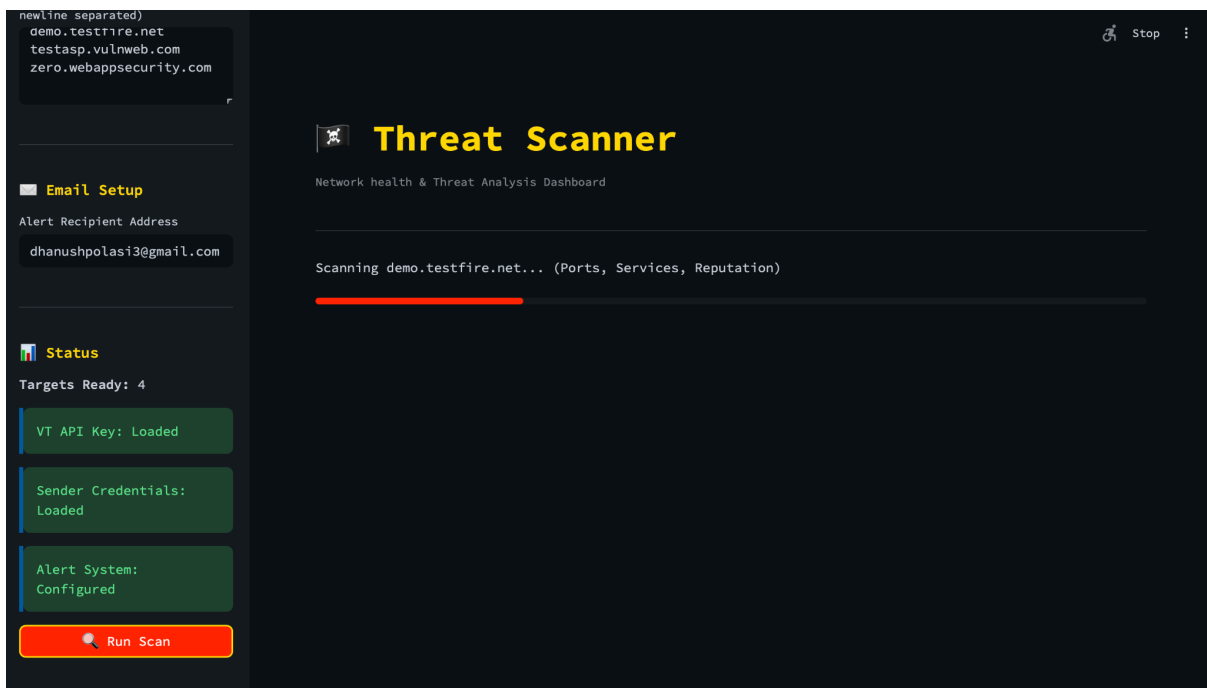
Screenshots of the output:



Before configuring the settings



After configuring settings



During the scan

zero.webappsecurity.com

Email Setup

Alert Recipient Address

dhanushpolasi3@gmail.com

Status

Targets Ready: 4

VT API Key: Loaded

Sender Credentials: Loaded

Alert System: Configured

Run Scan

Automated alert sent successfully!

Threat Scanner

Network health & Threat Analysis Dashboard

Scan Completed: 15 Mar 2026 17:48:30

Summary of the Scan

Target	Vulnerability	Severity	Score	Recommendation
testphp.vulnweb.com	Malicious Reputation (1 VT engines flagged)	Critical	10	Investigate immediately for comp
demo.testfire.net	Unencrypted Web Traffic (Port 80)	Medium	4	Enforce HTTPS/TLS and redirect
demo.testfire.net	Open Port 443 (https)	Low	2	Ensure service is required, pat
demo.testfire.net	Unencrypted Web Traffic (Port 8080)	Medium	4	Enforce HTTPS/TLS and redirect
demo.testfire.net	Open Port 8443 (https-alt)	Low	2	Ensure service is required, pat
testasp.vulnweb.com	Unencrypted Web Traffic (Port 80)	Medium	4	Enforce HTTPS/TLS and redirect
zero.webappsecurity.com	Unencrypted Web Traffic (Port 80)	Medium	4	Enforce HTTPS/TLS and redirect

As one of the identified vulnerability is CRITICAL, an email triggered automatically (image is taken during the second scan, so the time appears different compared to rest of the images)

newline separated)

demo.testfire.net

testasp.vulnweb.com

zero.webappsecurity.com

Email Setup

Alert Recipient Address

dhanushpolasi3@gmail.com

Status

Targets Ready: 4

VT API Key: Loaded

Sender Credentials: Loaded

Alert System: Configured

Run Scan

Scan Completed: 15 Mar 2026 17:39:54

Targets Scanned: 4

Total Vulnerabilities: 9

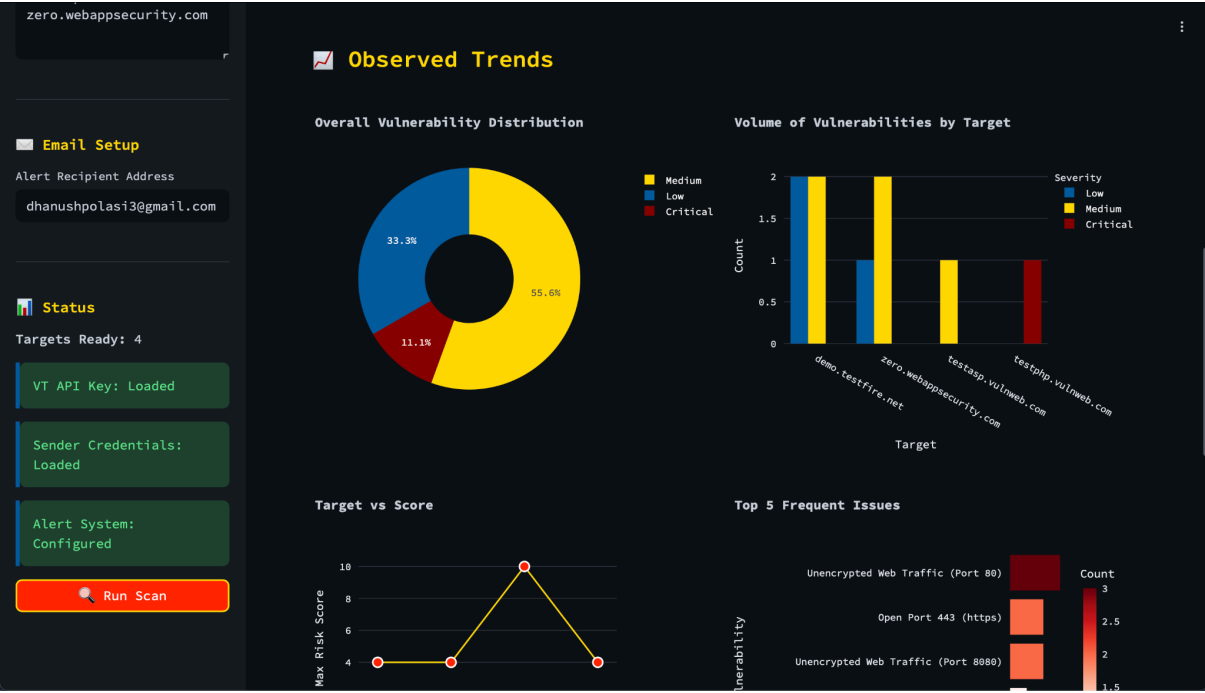
Critical/High Threats: 1

Max Risk Score: 10

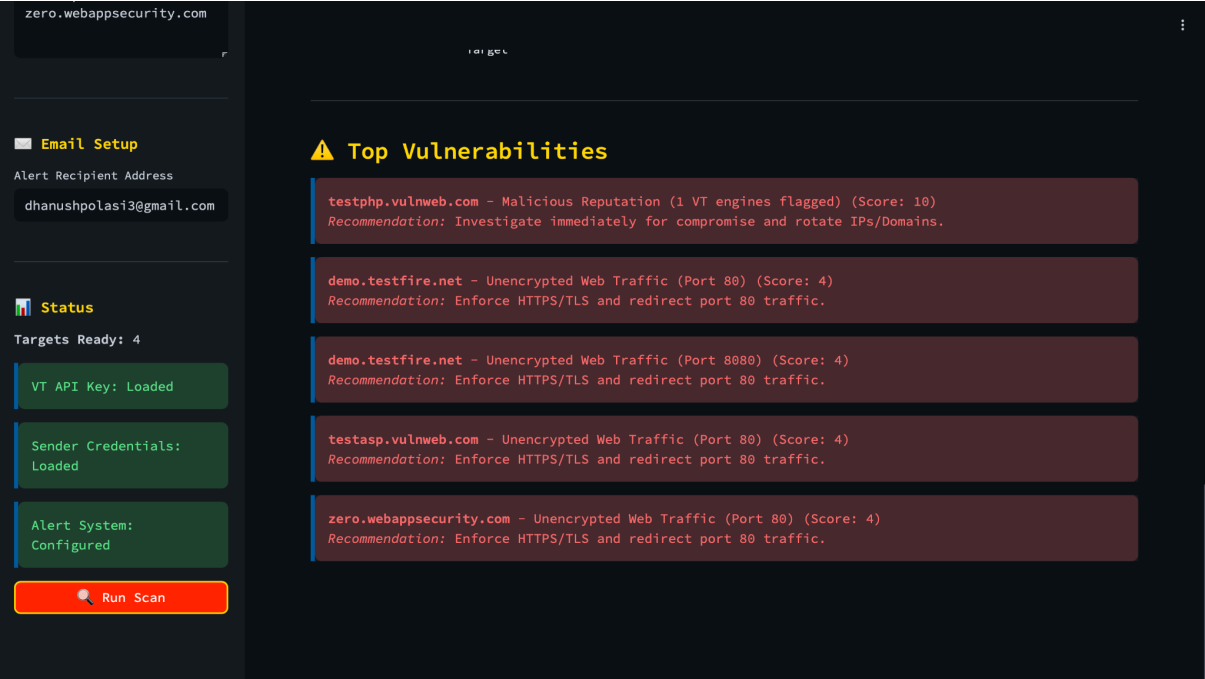
Summary of the Scan

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zero.webappsecurity.com	Unencrypted Web Traffic (Port 80)	Medium	4	Enforce HTTPS/TLS and redirect
zero.webappsecurity.com	Open Port 443 (https)	Low	2	Ensure service is required, pat
zero.webappsecurity.com	Unencrypted Web Traffic (Port 8080)	Medium	4	Enforce HTTPS/TLS and redirect

Summary of the Scan



Representation of Results



Vulnerabilities sorted by highest risk score

[CRITICAL] Vulnerability Alert for testphp.vulnweb.com Inbox x



dhanushpolasi3@gmail.com
to me ▾

Sun 15 Mar, 23:18 (12 days ago) ☆ 😊 ↩ ⋮

Threat Scanner - Security Alert

Scan Timestamp: 15 Mar 2026 17:48:30

Target(s): testphp.vulnweb.com

Overall Risk Score: **10/10**

High & Critical Findings Summary

Target	Vulnerability	Severity	Score	Recommended Action
testphp.vulnweb.com	Malicious Reputation (1 VT engines flagged)	Critical	10	Investigate immediately for compromise and rotate IPs/Domains.

Email Alert

Github README : [README.md](#)

DESIGN DECISIONS :

1. Not hardcoding the credentials in the code, and using colab secrets and env variables.
2. Instead of collecting the details of both sender and receiver for email alerts, we only collect receiver email, and configure the sender details by default. This enables actual user email reports.
3. As the number of targets is small, the data would be comparatively less, so instead of tabs, displayed the results in sequence.
4. Displaying the status of api key, and email system, so that the user would know whether the system is ready to use or not.
5. Adding a recommendation tab, to fix the vulnerability, helps the user to start the process of fixing the vulnerability.

CHALLENGES FACED:

1. As streamlit runs in its own environment in colab, i faced trouble in using the google colab secrets. Fixed it by storing the secrets into variables, and then passing them as env variables for the streamlit env.
2. Streamlit doesn't directly run on colab, so had to use cloudflare tunnel to run it.
3. Designing the html formatted email was tough, but referring to html fundamentals, helped in framing the email.
4. Emails should be triggered only when the alert is critical, so the scores of vulnerabilities were checked, and only the vulnerabilities with score ≥ 7

Dhanush P
#Z-13-11